WHY RUNNABLEBRANCH?

Sometimes we do not want to execute all paths.
We want to choose only one path based on a condition.

To handle this requirement, we use RunnableBranch.

WHAT IS RUNNABLEBRANCH?

RunnableBranch selects one runnable from multiple runnables based on conditions.

COMPARISON

RunnableSequence → Executes tasks one after another
RunnableParallel → Executes all branches at the same time
RunnableBranch → Executes only one branch based on condition

SYNTAX

```
RunnableBranch(
(condition1, runnable1),
(condition2, runnable2),
(condition3, runnable3),
default_runnable
)
```

IMPORTANT RULE

Only the first matching condition will be executed.
If a condition is satisfied, remaining conditions are not checked.

PROGRAM 1 – PASS OR FAIL

Logic:
marks ≥ 35 → PASS
else → FAIL

Output:
Input 70 → PASS
Input 20 → FAIL

PROGRAM 2 – IF-ELIF-ELSE

Logic:
marks ≥ 75 → Distinction
marks ≥ 60 → First Class
marks ≥ 35 → Pass
else → Fail

Output:
85 → Distinction
65 → First Class
45 → Pass
20 → Fail

PROGRAM 3 – DICTIONARY INPUT

Input:
{
"name": "Ramesh",
"marks": 70
}

Logic:
marks ≥ 35 → passed
else → failed

Output:
Ramesh passed the exam

PROGRAM 4 – AI MATERIAL GENERATOR

Requirement:
Generate study material based on level

Levels:
Beginner

Intermediate
Advanced

Logic:
If level is "beginner" → beginner_chain
If level is "intermediate" → intermediate_chain
Else → advanced_chain

Example Input:
{
"topic": "RunnableParallel",
"level": "beginner"
}

Output:
Beginner level study material

PROGRAM 5 – TRAVEL AGENT

Requirement:
Generate travel plan based on budget

Logic:
budget ≥ 100000 → Luxury plan
budget ≥ 50000 → Standard plan
else → Budget plan

Example Input:
{
"destination": "Goa",
"days": 5,
"budget": 75000
}

Output:
Standard travel plan

PROGRAM 6 – RECIPE GENERATOR (SWITCH CASE)

Requirement:
Act like switch-case

Logic:
food_type = "veg" → Vegetarian recipe
food_type = "non-veg" → Non-vegetarian recipe
else → General recipe

Examples:

Input:
{"food_type": "veg", "ingredient": "Paneer"}
Output: Vegetarian recipe

Input:
{"food_type": "non-veg", "ingredient": "Chicken"}
Output: Non-vegetarian recipe

Input:
{"food_type": "other", "ingredient": "Rice"}
Output: General recipe

KEY POINTS

- RunnableBranch executes only one branch
- Works like if / elif / else
- Can behave like switch-case
- First matched condition is executed
- Default runnable acts as fallback

FINAL SUMMARY

Use RunnableBranch when:

- You need decision-making logic
- Only one output is required
- Execution depends on conditions